MIT ACADEMY OF ENGINEERING, PUNE

Department of Computer Engineering | Academic Year 2026-27

COMPREHENSIVE VIVA VOCE & ORAL DEFENSE PREPARATION GUIDE

High-Impact Faculty Defense Q&A;, Mathematical Derivations & Rubric 5/5 Justifications

Q1: Explain why the deterministic CPM schedule is 54 working days while the academic semester is 16 weeks (80 working days).

Answer: The CPM calculations establish the pure deterministic minimum working schedule along the longest unyielding sequence of dependent tasks (54 working days across 13 zero-float activities). The difference of 26 working days (80 - 54) represents our scheduled operational buffer. This buffer absorbs mid-term examinations, institutional project reviews, and pilot school holidays without slipping our target delivery date.

Q2: At Day 35, your EVM report shows CPI = 1.050 and SPI = 0.928. How do you interpret this project state to the steering committee?

Answer: The project is in a highly favorable cost position (CV = +INR 14,143, CPI = 1.050), meaning we get INR 1.05 of physical progress for every rupee spent. The slight schedule delay (SV = -INR 23,000, SPI = 0.928) is isolated in ACT-15 (Document OCR Verification under CA-03). Because ACT-15 has 4 days of total float on a non-critical path, this delay does not affect the critical path. The forecasted completion cost EAC is INR 3,90,476, yielding a projected cost surplus of INR 19,524.

Q3: How do you differentiate between Contingency Reserve and Management Reserve in your cost model?

Answer: Contingency Reserve (10% / INR 41,000) addresses 'known-unknowns' quantitatively identified in our Risk Register (RSK-01 to RSK-10). It is under the direct spending authority of the Project Manager. Management Reserve (7% / INR 34,000) addresses 'unknown-unknowns' or major policy/regulatory shifts and is controlled strictly by the MITAOE Project Sponsor, requiring formal Change Control Board (CCB) approval.

Q4: How does Little's Law prevent bottlenecks in your Kanban Board?

Answer: Little's Law states that WIP = Throughput (lambda) x Lead Time (W). By capping WIP to 3 tasks in 'In Progress' and 3 in 'Testing' (total flight WIP <= 6), we eliminate context switching and queuing bottlenecks. This increases throughput from 0.25 to 0.37 tasks/day, compressing average cycle time from 34.3 days down to 16.2 days per task (a 52.8% reduction).

Q5: How do you defend your schedule compression model if the sponsor demands a 4-day delivery acceleration?

Answer: We apply our mathematical Crashing Cost Slope formula: Cost Slope = (Crash Cost - Normal Cost) / (Normal Time - Crash Time). We select ACT-06 (Spring Boot Setup, cost slope INR 3,000/day) and ACT-17 (Playwright Integration, cost slope INR 2,500/day), crashing each by 2 days. This compresses the critical path from 54 to 50 days at an additional cost of INR 11,000, funded from our favorable cost surplus (INR 14,143) without exceeding the original BAC budget.